

Multi-Class Disease Diagnosis Using Multinomial Logistic Regression

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Abstract

This paper presents a machine-learning framework for multi-class disease diagnosis using patient symptoms and profile information. The supplied healthcare project identifies fever, cough, fatigue, difficulty breathing, age, gender, blood pressure, and cholesterol level as representative input features and selects Multinomial Logistic Regression for disease classification. The workflow includes data preprocessing, categorical encoding, feature preparation, train-test splitting, model training, disease prediction, and evaluation using accuracy, precision, recall, and F1-score.

Keywords: Multinomial Logistic Regression; Disease Classification; Healthcare; Machine Learning; Patient Symptoms; Decision Support

1. Introduction

Disease diagnosis requires analysis of multiple symptoms and patient characteristics. Similar symptoms may occur across different diseases, making classification challenging when records are reviewed manually. The healthcare project proposes a machine-learning decision-support approach in which patient symptoms and profile information are transformed into a predicted disease category.

The project identifies fever, cough, fatigue, difficulty breathing, age, gender, blood pressure, and cholesterol level as representative input features and selects Multinomial Logistic Regression for multi-class disease classification.

1.1 Problem Statement

- Disease diagnosis involves multiple symptoms and patient characteristics.
- Different diseases can share similar symptoms.
- Manual analysis of large patient records can be time-consuming.
- Machine learning can provide a preliminary prediction of the most likely disease category.

1.2 Objectives

- Develop a multi-class disease prediction model.
- Use symptoms and patient profile information as input features.
- Apply Multinomial Logistic Regression for disease classification.
- Evaluate the model using accuracy, precision, recall, and F1-score.

2. Related Work

The supplied reference papers demonstrate how statistical methods can transform complex information into interpretable analytical representations. The first reference is a systematic review of Principal Component Analysis in soccer, basketball, and rugby, while the second presents a scalable PCA framework for NBA player and team comparisons. These studies motivate structured statistical learning for complex datasets, although the present healthcare project uses Multinomial Logistic Regression as the primary classifier.

3. Dataset and Exploratory Data Analysis

The healthcare project describes a Disease Symptoms & Patient Profile Dataset. The documented EDA includes checking dataset structure, identifying missing values and duplicates, analyzing disease/class distribution, studying age and gender distribution, and examining relationships between input features.

Feature group	Variables / analysis
Symptoms	Fever, cough, fatigue, difficulty breathing
Patient profile	Age, gender
Clinical profile	Blood pressure, cholesterol level
Data quality	Missing values and duplicates
Distribution	Disease/class, age, and gender distributions
Relationships	Relationships among input features

Table 1. Dataset features and exploratory analysis.

4. Method

4.1 Data Preprocessing

The proposed workflow cleans the data and converts categorical values into numerical form. Selected features are prepared for model training.

4.2 Multinomial Logistic Regression

For K disease classes, Multinomial Logistic Regression estimates a probability for each class. The predicted category is the class with the highest estimated probability.

4.3 Prediction Workflow

Stage	Process	Output
1	Patient data collection	Structured patient records
2	Data preprocessing	Cleaned and encoded data
3	Feature preparation	Model-ready feature matrix
4	Train-test split	Training and testing subsets
5	Model training	Fitted classifier
6	Prediction	Disease category
7	Evaluation	Accuracy, precision, recall, F1-score

Table 2. Disease prediction workflow.

5. Proposed Solution

The proposed system follows Patient Data Collection → Data Preprocessing → Feature Preparation → Model Training → Disease Prediction. The trained model estimates the most likely disease category from patient symptoms and profile information.

6. Results and Discussion

The supplied project specifies accuracy, precision, recall, and F1-score as evaluation measures but does not provide numerical experimental results. Therefore, no unsupported accuracy value is reported. The supported outcome is a structured data-driven method for preliminary disease-category prediction.

7. Limitations

- The complete raw dataset is not included in the supplied project files.
- The supplied material does not report numerical model performance.
- The model is intended for preliminary decision support rather than autonomous diagnosis.

8. Conclusion

A multi-class disease prediction framework using Multinomial Logistic Regression was developed from the healthcare project requirements. Patient symptoms and profile information form the input, while the model produces the most likely disease category.

9. Future Scope

- Add more symptoms and medical features.
- Use larger and more diverse datasets.
- Compare with other machine-learning algorithms.

- Develop a web or mobile prototype.
- Explore advanced machine-learning and deep-learning techniques.

10. Author Contributions and Declarations

Conceptualization, methodology, analysis, and writing: M. Geethika Chowdary and G. Varshini. Academic guidance: Shaik Asif. Funding and conflicts of interest were not specified in the supplied project files.

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